

Section 7.4

Exercise (7.4.7). Let R be a commutative ring with 1. Prove that the principal ideal generated by x in the polynomial ring $R[x]$ is a prime ideal if and only if R is an integral domain. Prove that (x) is a maximal ideal if and only if R is a field.

Proof: Let R be a commutative ring with 1. Consider the homomorphism $\varphi : R[x] \rightarrow R$ defined by $\varphi(a_0 + a_1x + a_2x^2 + \dots) = a_0$. Let $p(x), q(x) \in R[x]$ such that $\varphi(p(x)) = a_0$ and $\varphi(q(x)) = b_0$. Note that by polynomial multiplication $\varphi(p(x)q(x)) = a_0b_0 = \varphi(p(x))\varphi(q(x))$ and clearly φ is a surjection. Also note that every $p(x) \in \ker(\varphi)$ must have zero constant term and therefore is comprised of products and sums of x so $\ker(\varphi) = (x)$.

By The First Isomorphism Theorem for Rings we find that $R[x]/(x) \cong R$. By Proposition 13, supposing (x) is a prime ideal in $R[x]$ we know that $R[x]/(x)$ is an integral domain, thus R is an integral domain. Since Proposition 13 is an if and only if statement if we suppose R is an integral domain then we get that $R/(x)$ is an integral domain and therefore (x) is a prime ideal.

Supposing (x) is a maximal ideal we get that $R/(x)$ is a field by Proposition 12 and therefore R is a field. Again since Proposition 12 is an if and only if statement, supposing R is a field we know that $R/(x)$ is a field and therefore (x) must be a maximal ideal.

□

Exercise (7.4.30). Let I be the ideal of the commutative ring R and define,

$$\text{rad } I = \{r \in R \mid r^n \in I \text{ for some } n \in \mathbb{Z}^+\}$$

called the radicals of I . Prove that $\text{rad } I$ is an ideal containing I and that $\text{rad } I/I$ is the nilradical of quotient ring R/I .

Proof: Suppose I is an ideal of the commutative ring R and $\text{rad } I$ is defined as above. Clearly

$I \subseteq \text{rad } I$, let $i \in I$ and note that $i^1 \in I$ so $i \in \text{rad } I$. Let $a, b \in \text{rad } I$, by definition there exists some $i, j \in \mathbb{Z}^+$ such that $a^i, b^j \in I$. Note that by binomial expansion we

know that

$$(a - b)^{i+j} = \sum_{k=0}^{i+j} \binom{i+j}{k} a^{i+j-k} (-b)^k.$$

When $k \geq j$, since R is commutative we can factor a b^j term and we conclude that every term in the sum where $k \geq j$ is in I . When $k < j$, since R is commutative we can factor an a^i term and we conclude that every term in the sum where $k < j$ is in I . Thus every term in the sum is in I and $(a - b)^{i+j} \in I$. Thus by definition $(a - b) \in \text{rad } I$.

Since R is commutative $(ab)^{ij} = a^{ij}b^{ij} = a^{ij}(b^j)^i \in I$ so $ab \in \text{rad } I$. So $\text{rad } I$ is a subring of R .

Suppose $r \in R$ and consider that since R is commutative $(ar)^i = a^i r^i \in I$. Therefore $ar \in \text{rad } I$, since I is an ideal the same argument can be applied for $ra \in \text{rad } I$. Thus $\text{rad } I$ is an ideal in R .

The nilradical of the quotient ring R/I is the ideal containing all the nilpotent elements of R/I . By definition if an element $(x + I) \in R/I$ is nilpotent then there exists some n such that $(x + I)^n = x^n + I = I$ (well defined since R is commutative, and recall that I is the zero in R/I). Thus $x^n \in \text{rad } I$ and we conclude that $\text{rad } I/I$ is the nilradical of quotient ring R/I . $\square$

Exercise (7.4.41). A proper ideal Q of the commutative ring R is called primary if whenever $ab \in Q$ and $a \notin Q$ then $b^n \in Q$ for some positive integer n . Establish the following facts about primary ideals,

- (a) The primary ideals of $\mathbb{Z}$ are 0 and (p^n) , where p is a prime and n is a positive integer.

Proof: Clearly 0 is a primary ideal of $\mathbb{Z}$. Let $ab \in (p^n)$ then either $p \mid a$ or $p \mid b$ which implies either $a^n \in (p^n)$ or $b^n \in (p^n)$. Suppose for the sake of contradiction (i) is a prime ideal where i is not prime or prime power. Let $i = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_n^{\alpha_n}$ where p_k are prime and note that by definition $(p_1^{\alpha_1} p_2^{\alpha_2} \dots p_{n-1}^{\alpha_{n-1}})(p_n^{\alpha_n}) = i \in (i)$ however clearly neither $p_n^{\alpha_n} \in (i)$ or $(p_1^{\alpha_1} p_2^{\alpha_2} \dots p_{n-1}^{\alpha_{n-1}}) \in (i)$ so (i) is not primary. $\square$

- (b) Every prime ideal of R is a primary ideal.

Proof: Suppose P is a prime ideal. By definition if $ab \in P$ then either $a \in P$ or $b \in P$. Without loss of generality suppose $a \notin P$ and $b \in P$, then clearly $b^n \in P$ and we find that P is primary. $\square$

- (c) An ideal Q of R is primary if and only if every zero divisor in R/Q is a nilpotent element of R/Q .

Proof: Suppose Q , an ideal of R is primary. Let $(i + Q) \in R/Q$ be a zero divisor, so $(j + Q) \in R/Q$ such that $(i + Q)(j + Q) = (ij + Q) = Q$ where $(i + Q), (j + Q) \neq Q$. Therefore $ij \in Q$ and $j \notin Q$ since Q is primary, there exists some $i^n \in Q$ so we get $(i + Q)^n = (i^n + Q) = Q$. Thus $(i + Q)$ is nilpotent in R/Q .

Suppose every zero divisor in R/Q is a nilpotent element of R/Q . Let $ab \in Q$ and $a \notin Q$ note that $(ab + Q) = (a + Q)(b + Q) = (a + Q)(Q) = Q$. Thus $(b + Q)$ is a zero divisor. Then there exists some $(b + Q)^n = (b^n + Q) = Q$ so $b^n \in Q$. Therefore Q is primary. $\square$

- (d) If Q is a primary ideal then $\text{rad}(Q)$ is a prime ideal.

Proof: Suppose Q is a primary ideal. Let $ab \in \text{rad}(Q)$. By definition $\text{rad}(Q)$ there exists some n such that $(ab)^n \in Q$. Since R is commutative we know that $(a^n)(b^n) \in Q$, without loss of generality suppose $(a^n) \notin Q$, since Q is primary then there exists some $b^{mn} \in Q$. Therefore $b \in \text{rad}(Q)$. $\square$

Section 9.1

Exercise (9.1.1). Let $p(x, y, z) = 2x^2y - 3xy^3z + 4y^2z^5$ and $q(x, y, z) = 7x^2 + 5x^2y^3z^4 + 3x^2z^3$ be polynomials in $\mathbb{Z}[x, y, z]$.

- (a) Write each of p and q as polynomials in x with coefficients in $\mathbb{Z}[y, z]$

Proof: Writing p and q as polynomials in x with coefficients in $\mathbb{Z}[y, z]$ we get,

$$p(x) = (4y^2z^5) - (3y^3z)x + (2y)x^2,$$

$$q(x) = (7 + 5y^3z^4 + 3z^3)x^2.$$

□

- (b) Find the degree of each of p and q .

Proof: By definition we know that the degree of a nonzero polynomial is the largest degree of any of its monomial terms and the degree of a monomial $x_1^{d_1}x_2^{d_2}x_3^{d_3}\dots x_n^{d_n}$ is simply $d_1 + d_2 + d_3 + \dots + d_n$. Thus $\deg(p) = 7$ and $\deg(q) = 9$. □

- (c) Find the degree of p and q in each of the three variables x, y , and z

Proof: As we have shown in part a, $\deg_x(p) = 2$ and $\deg_x(q) = 2$. Note that p and q as polynomials in y with coefficients in $\mathbb{Z}[x, z]$ we get,

$$p(y) = (2x^2)y + (4z^5)y^2 - (3zx)y^3,$$

$$q(y) = (7x^2 + 3x^2z^3) + (5x^2z^4)y^3.$$

So $\deg_y(p) = 3$ and $\deg_y(q) = 3$. Note that p and q as polynomials in z with coefficients in $\mathbb{Z}[x, y]$ we get,

$$p(y) = 2x^2y - (3xy^3)z + (4y^2)z^5,$$

$$q(y) = 7x^2 + (3x^2)z^3 + (5x^2y^3)z^4.$$

So $\deg_z(p) = 5$ and $\deg_z(q) = 4$. □

- (d) Compute pq and find the degree of pq in each of the three variables x, y , and z .

Proof: Computing pq we get,

$$\begin{aligned}
 pq &= (2x^2y - 3xy^3z + 4y^2z^5)(7x^2 + 5x^2y^3z^4 + 3x^2z^3) \\
 &= (2x^2y - 3xy^3z + 4y^2z^5)7x^2 + (2x^2y - 3xy^3z + 4y^2z^5)5x^2y^3z^4 + (2x^2y - 3xy^3z + 4y^2z^5)3x^2z^3 \\
 &= 2x^2y(7x^2) - 3xy^3z(7x^2) + 4y^2z^5(7x^2) + 2x^2y(5x^2y^3z^4) - 3xy^3z(5x^2y^3z^4) \\
 &\quad + 4y^2z^5(5x^2y^3z^4) + 2x^2y(3x^2z^3) - 3xy^3z(3x^2z^3) + 4y^2z^5(3x^2z^3) \\
 &= 14x^4y - 21x^3y^3z + 28y^2z^5x^2 + 10x^4y^4z^4 - 15x^3y^6z^5 + 20x^2y^5z^9 + 6x^4yz^3 - 9x^3y^3z^4 + 12y^2z^8x^2
 \end{aligned}$$

Note that $\deg(pq) = \deg(p) + \deg(q) = 16$. Finding the degree of pq in each variable we get $\deg_x(pq) = 4$, $\deg_y(pq) = 6$, and $\deg_z(pq) = 9$. $\square$

(e) Write pq as a polynomial in the variable z with coefficients in $\mathbb{Z}[x, y]$

Proof: Computing pq as a polynomial in the variable z with coefficients in $\mathbb{Z}[x, y]$ we get,

$$pq(z) = 14x^4y - (21x^3y^3)z + (6x^4y)z^3 + (10x^4y^4 - 9x^3y^3)z^4 + (28y^2x^2 - 15x^3y^6)z^5 + (12y^2x^2)z^8 + (20x^2y^5)z^9.$$

$\square$

Exercise. 9.1.4 Prove that the ideals (x) and (x, y) are prime ideals in $\mathbb{Q}[x, y]$ but only the latter ideal is a maximal ideal.

Proof: Suppose $p, q \in \mathbb{Q}[x, y]$ such that $pq \in (x)$. Writing p and q as polynomials in the variable x with coefficients in $\mathbb{Q}[y]$ we get,

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

$$q(x) = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m$$

Now consider pq and note that a_0b_0 is a term in pq which is degree 0 in the variable x , however since $pq \in (x)$ it must be the case that $a_0b_0 = 0$. By Proposition 1 since

$\mathbb{Q}$ is an integral domain we know that $\mathbb{Q}[y]$ is an integral domain so it must follow that either $a_0 = 0$ or $b_0 = 0$. Without loss of generality suppose $a_0 = 0$, then we can factor an x out of p and we conclude that $p \in (x)$.

Suppose $p, q \in \mathbb{Q}[x, y]$ such that $pq \in (x, y)$. Again it must follow that the constant terms in a_0 and b_0 in p and q must have the property that $a_0b_0 = 0$. By Proposition 1 since $\mathbb{Q}$ is an integral domain we know that $\mathbb{Q}[x, y]$ is an integral domain so it must follow that either $a_0 = 0$ or $b_0 = 0$. Without loss of generality suppose $a_0 = 0$, then every term $p \in (x, y)$.

Apply Exercise 7.4.7 to show that (x, y) is maximal in $\mathbb{Q}(x, y)$ by noting that $\mathbb{Q}$ is a field. □

Exercise. 9.1.9 Prove that a polynomial ring in infinitely many variables with coefficients in any commutative ring contains ideals that are not finitely generated.

Proof: Suppose R is a commutative ring and $R[x_1, x_2, \dots]$ and for the sake of contradiction suppose $I = (x_1, x_2, \dots, x_n)$ is an ideal. Let $x_m \in R[x_1, x_2, \dots]$ such that $m > n$, and note that since I is an ideal then $x_m I \in I$, however by definition clearly any monomial with x_m is not in I . □